AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 18 · WORKBOOK

Confidence Is Not Correctness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Confidence Is Not Correctness before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Prediction Is Not Fact

Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Save New Examples for Testing

Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Debug the Cause

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Confidence Is Not Correctness trace and record the first step you can explain without help.

Use W2: Judge cases using outcomes for **Confidence** and **correctness**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Confidence Is Not Correctness explanation against the supplied facts before discussing it.

Use the worked case for **Confidence** and **correctness**. Proposed explanation: “An 80 point score always means an 80 percent probability”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Audit a different probability claim

A fictional leaf sorter claims a 75 percent probability of **correctness** for each of eight chosen labels. Later **evidence** establishes six correct labels, one incorrect label and one missing outcome.

A report says six out of eight proves perfect **calibration** and guarantees the next label. Repair the report, state the assessed and unassessed counts, and describe a better follow-up check.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Confidence Is Not Correctness target to a trace and a justified conclusion.

A helper says 'definitely correct' on four answers; only two match the answer key. A cautious helper gives the same four answers. Which is more accurate? What additional **evidence** would you need to compare **confidence** quality?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Confidence Is Not Correctness case and show the deciding evidence.

A helper confidently labels a blurred dog photo as cat; the clear original shows a dog.

What does this show?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A sorter is very confident that a leaf is a coin, but a person checks and sees a leaf. Which determines **correctness**: **confidence** or the checked object? Why?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 18 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Confidence Is Not Correctness adds to Helpful Classroom Sorter and keep the

> evidence.

Record both the sorter's **confidence** wording and the human-confirmed outcome; keep them in separate columns.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A helper says 'definitely correct' on four answers; only two match the answer key. A cautious helper gives the same four answers. Which is more accurate? What additional **evidence** would you need to compare **confidence** quality?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Read the score definition

> Read the score definition. Apply the supplied Confidence Is Not Correctness facts and keep your reasoning.

A card sorter gives a match score of 80 out of 100 points. Its documentation says the score counts weighted feature matches and is not a probability. A learner calls this an 80 percent chance of **correctness**.

Explain the error and write an accurate description.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Judge cases using outcomes

> Judge cases using outcomes. Apply the supplied Confidence Is Not Correctness facts and keep your reasoning.

Use cases A, B and C from the lesson table. State which is correct, incorrect and unassessed. Explain why ordering them by **confidence** cannot replace checking their outcomes.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Interpret a small group carefully

> Interpret a small group carefully. Apply the supplied Confidence Is Not Correctness facts and keep your reasoning.

Ten **prediction**s each claim an 80 percent probability that their chosen category is correct. Five later prove correct and five incorrect. State the claimed probability and observed fraction separately.

Does this prove every future **prediction** will be wrong, or that each next **prediction** has exactly a 50 percent chance? Explain.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Audit a different probability claim

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Audit a different probability claim. Apply the supplied Confidence Is Not Correctness facts and keep your reasoning.

A fictional leaf sorter claims a 75 percent probability of **correctness** for each of eight chosen labels. Later **evidence** establishes six correct labels, one incorrect label and one missing outcome.

A report says six out of eight proves perfect **calibration** and guarantees the next label. Repair the report, state the assessed and unassessed counts, and describe a better follow-up check.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Keep repetition separate from evidence

> Keep repetition separate from evidence. Apply the supplied Confidence Is Not Correctness facts and keep your reasoning.

A learner copies the same 90 percent **prediction** into ten rows and calls this ten independent tests. No new outcomes were obtained. Explain why the **evidence** has not grown and propose a useful check.

> **My reasoning and evidence**

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Confidence Is Not Correctness

> A fictional card classifier reports a claimed 80% chance that each chosen label is correct. Explain the labelled step before checking the changed condition.

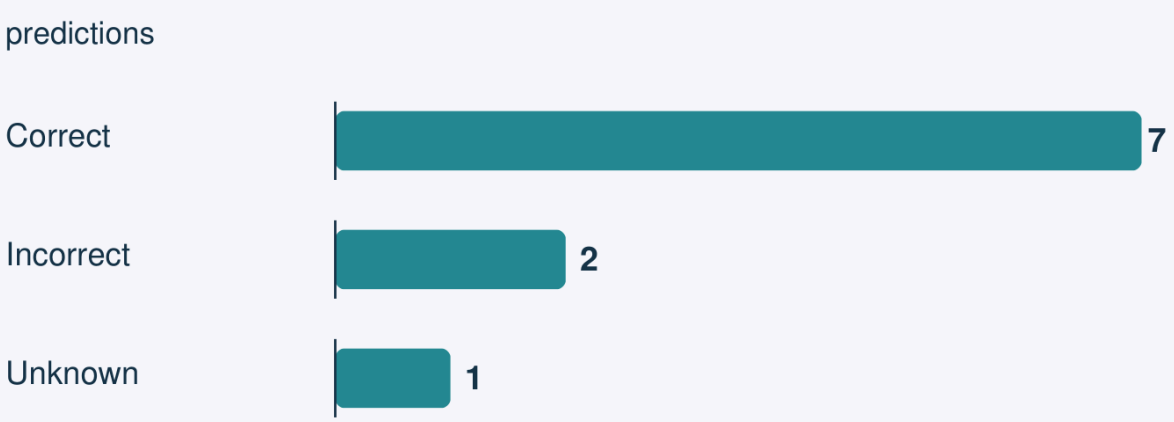
A fictional card classifier reports a claimed **80%** chance that each chosen label is correct.

Ten **prediction**s all have score **80%**.

Later checks find 7 correct, 2 incorrect and 1 outcome unavailable. The selected cards were not a representative survey.

AI ACADEMY / AI-1/18

Known outcomes and missing evidence



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/18

Confidence Is Not Correctness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Keep scores

Ten chosen labels each carry 80%; this is a claim about uncertainty.

2. Checked outcomes

Seven correct plus two wrong gives nine known outcomes.

3. Report coverage

Observed agreement is 7/9 among checked cards; outcome coverage is 9/10.

4. Retain unknown

The missing result could change the all-ten correct count to 7 or 8.

CHANGE ONE THING

The missing outcome is checked and found incorrect.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Was the unknown label guaranteed correct because its **confidence** was **80%**?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Keep sorter scores, reference outcomes and missingness separately; use review policies instead of treating high scores as facts.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Keep sorter scores, reference outcomes and missingness separately; use review policies instead of treating high scores as facts.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional card classifier reports a claimed **80%** chance that each chosen label is correct.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The missing result could change the all-ten correct count to 7 or 8.

1. Checked outcomes

2. Report coverage

3. Retain unknown

4. Keep scores

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Retain unknown The missing result could change the all-ten correct count to 7 or 8.

Do not count the missing outcome as correct or silently remove it from the coverage report. These records allow an all-ten agreement range from **70%** to **80%**.

They do not show that claimed probabilities match checked **correctness** over many comparable **prediction**s; that requires much more suitable **evidence**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The missing outcome is checked and found incorrect. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. All-ten agreement is 7/10=**70%**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: All-ten agreement is

7/10=**70%**. There are now three wrong labels and no unknown outcomes.

Repair a missing building block

Review **Prediction** Is Not Fact (ai-1/17). Goal: Separate a recorded **prediction** from the **observation** used to check it. Short practice: Which field changes the **observation** label, and which field changes the model **prediction**?.

Return to this week and explain what you changed.

Review Save New Examples for Testing (ai-1/15). Goal: Separate examples used to choose a model from examples reserved for its final evaluation. Short practice: Does 4/4 on the replay replace the original 3/4 result? Explain..

Return to this week and explain what you changed.

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose **input** 0 and give both predicted **output**s without inventing an observed result..

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

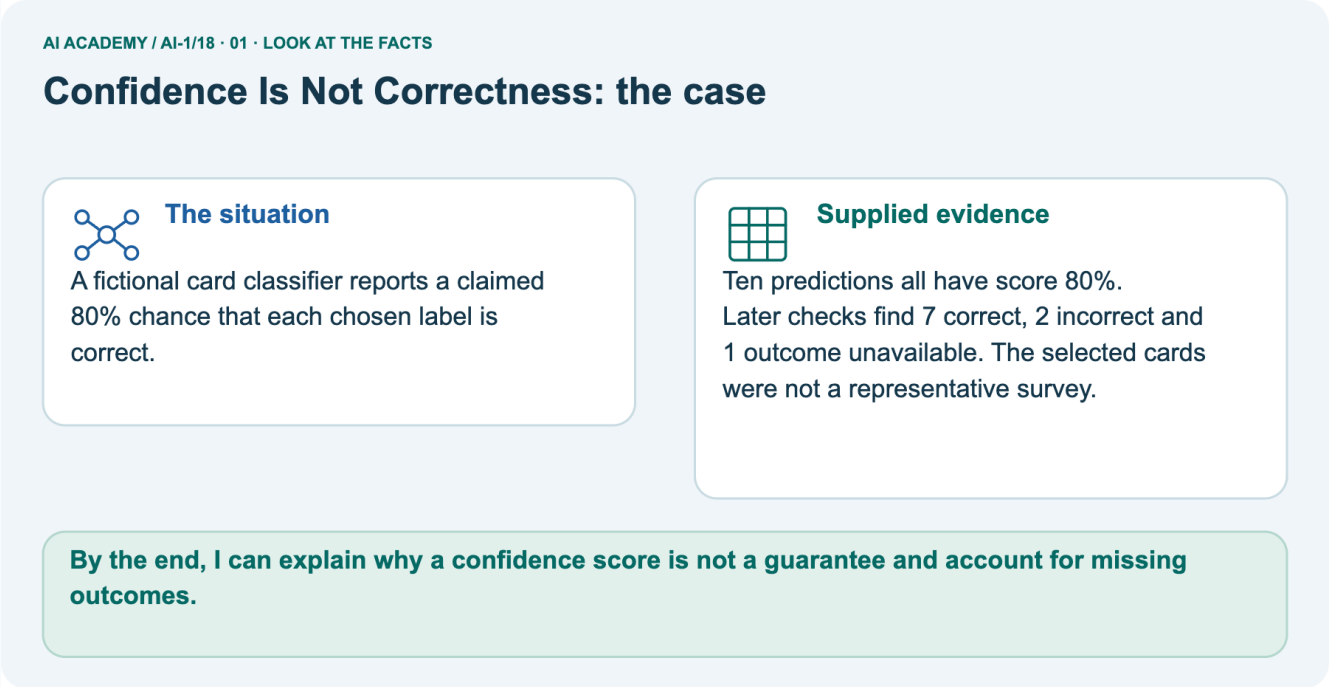
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Confidence Is Not Correctness picture from its labels, then record a relevant limit.

01. Confidence Is Not Correctness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

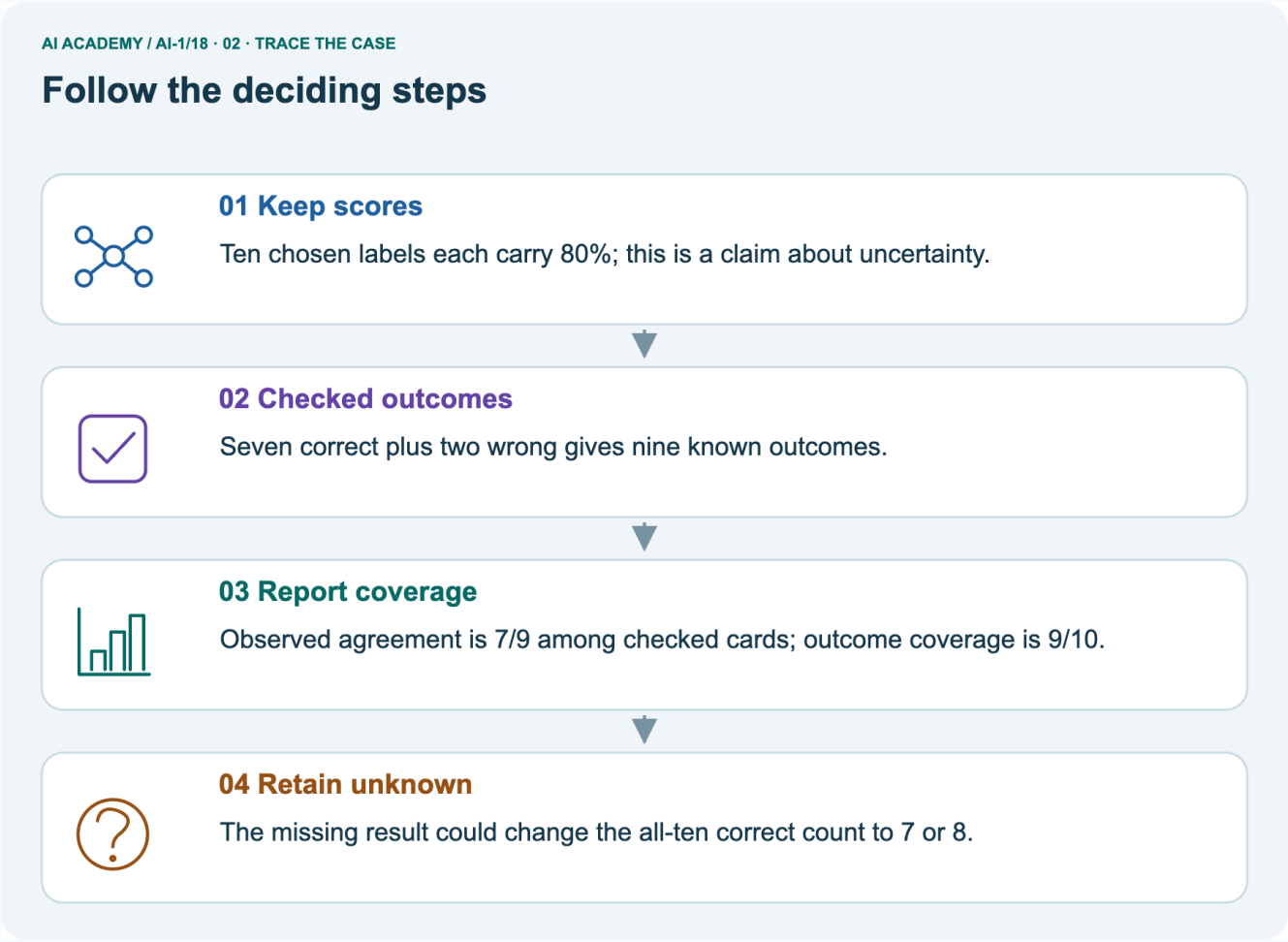
A fictional card classifier reports a claimed **80%** chance that each chosen label is correct. Supplied facts: Ten **prediction**s all have score **80%**. Later checks find 7 correct, 2 incorrect and 1 outcome unavailable.

The selected cards were not a representative survey.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep scores: Ten chosen labels each carry **80%**; this is a claim about uncertainty. → Checked outcomes: Seven correct plus two wrong gives nine known outcomes. → Report coverage: Observed agreement is 7/9 among checked cards; outcome coverage is 9/10. → Retain unknown: The missing result could change the all-ten correct count to 7 or 8.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Known outcomes and missing evidence



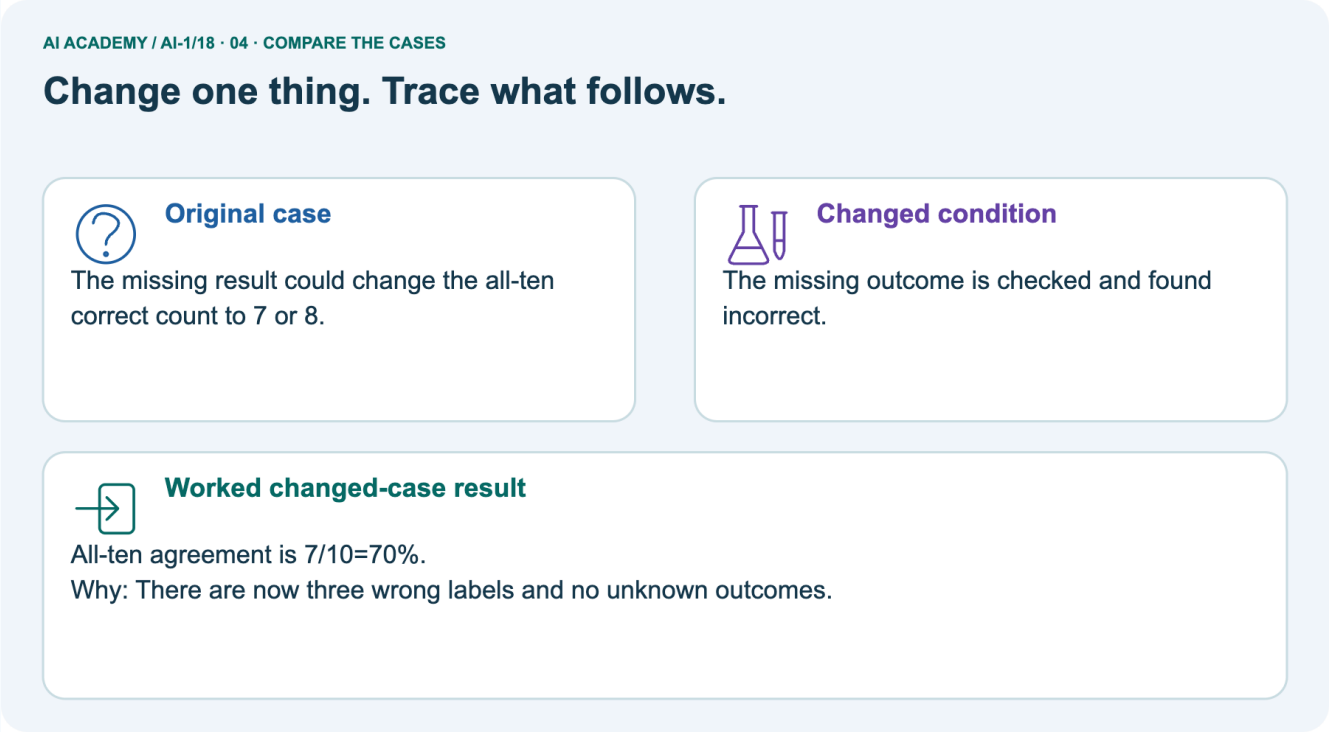
Follow the labelled picture with the matching explanation. Open the vector diagram

Known outcomes and missing **evidence**. Units: **prediction**s. Correct: 7. Incorrect: 2. Unknown: 1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

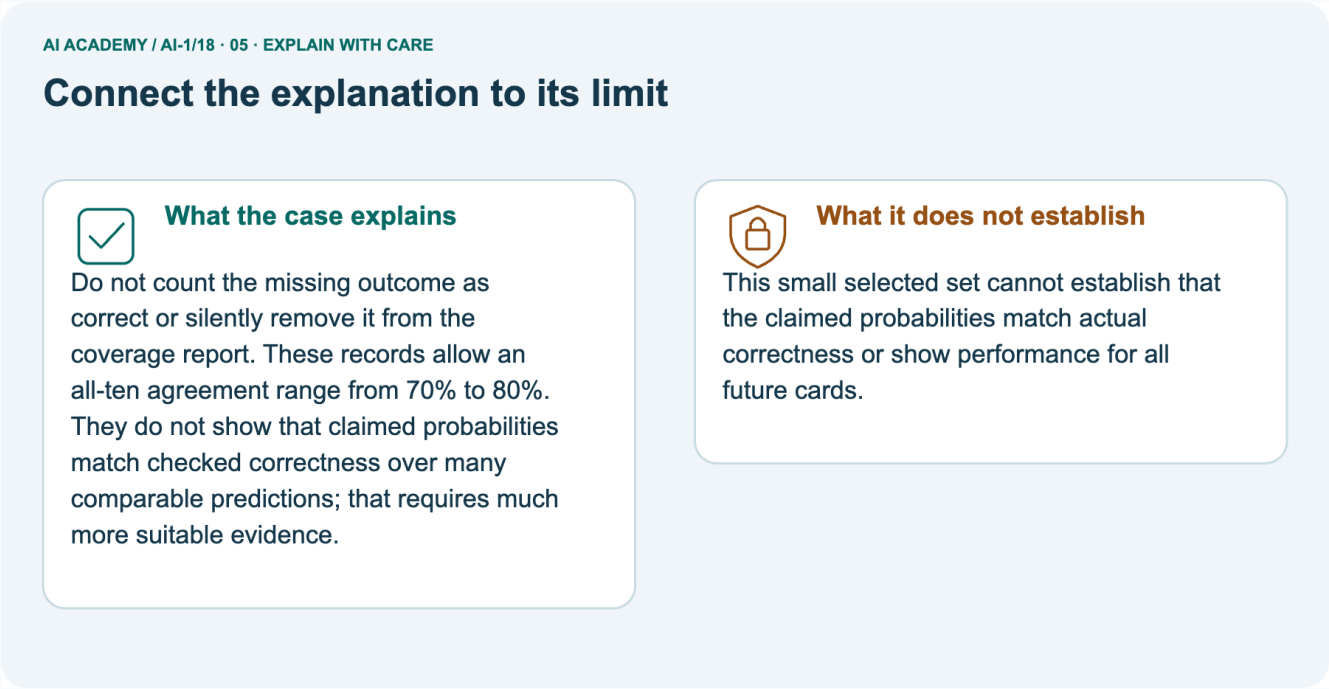
Original case: The missing result could change the all-ten correct count to 7 or 8. Changed condition: The missing outcome is checked and found incorrect. Expected result: All-ten agreement is 7/10=**70%**.

Reason: There are now three wrong labels and no unknown outcomes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Do not count the missing outcome as correct or silently remove it from the coverage report. These records allow an all-ten agreement range from **70%** to **80%**.

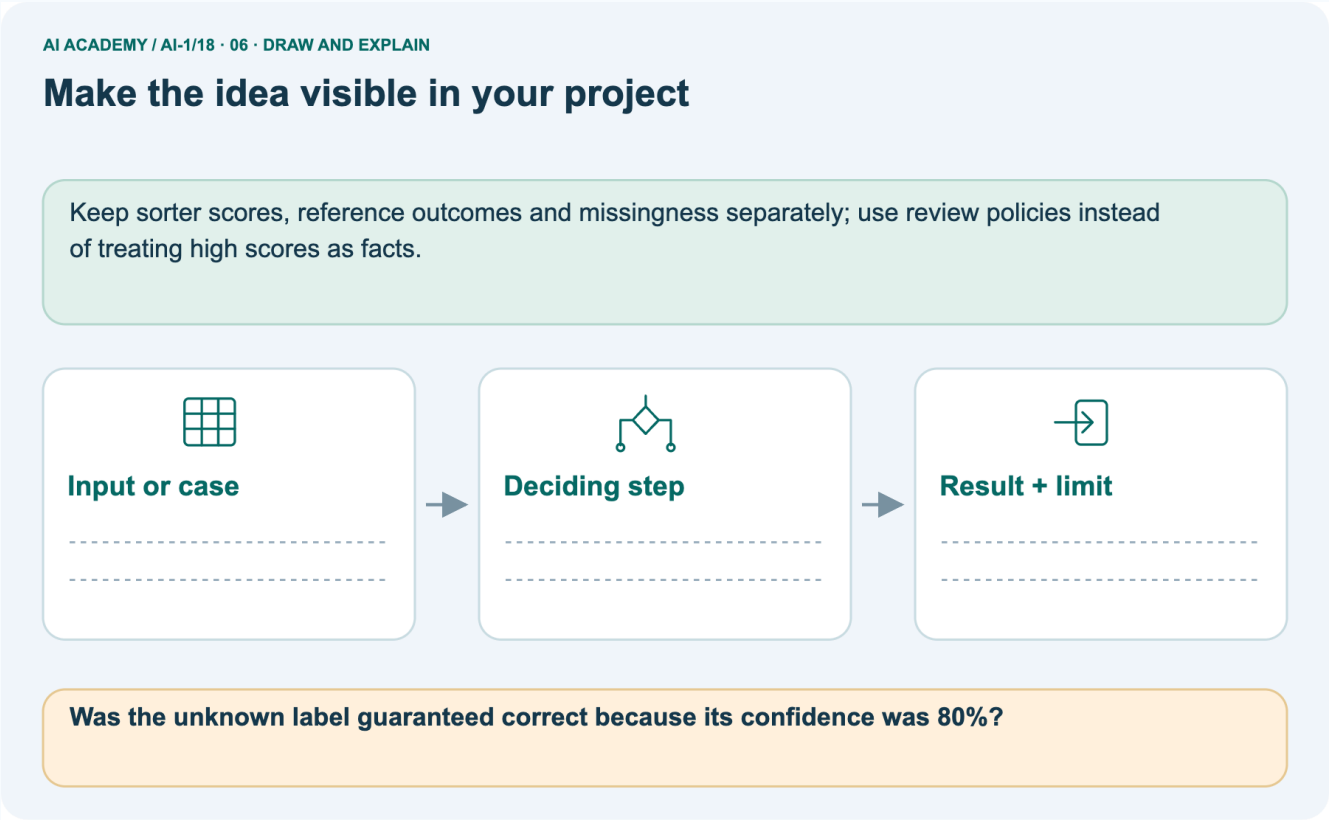
They do not show that claimed probabilities match checked **correctness** over many comparable **prediction**s; that requires much more suitable **evidence**.

Limit: This small selected set cannot establish that the claimed probabilities match actual **correctness** or show performance for all future cards.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep sorter scores, reference outcomes and missingness separately; use review policies instead of treating high scores as facts. Draw your **input**, deciding step, result and limit.

Supported question: Was the unknown label guaranteed correct because its **confidence** was **80%**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.